

Shayan Ghiasvand

ShayanGhiasvand

shayan-ghiasvand

✉ shayan.gh@hotmail.com

📍 Toronto, ON

☎ (438) 408-6373

EDUCATION

- **Concordia University** Montréal, Canada
Master of Applied Science in Mechanical Engineering May 2024
Thesis: Image Moment-based Visual Servoing for Satellite Target Tracking using a Robotic Manipulator
Courses: Autonomous Robotics Systems — Industrial Automation — Industrial Control Systems — Adaptive Control
- **University of Tehran** Tehran, Iran
Bachelor of Science in Mechanical Engineering May 2019
Thesis: Implementation of a Computer-based Control for Ball & Beam System
Courses: Mechatronics — Automatic Control

PUBLICATIONS

- **Ghiasvand, S.,** Xie, W.-F., & Mohebbi, A. (2024). "Deep neural network-based robotic visual servoing for satellite target tracking." *Frontiers in Robotics and AI*

EXPERIENCE

- **University of Toronto** Toronto, Canada
[Robotics Engineer](#) Jul 2024 - Oct 2024
Robotized an RC excavator by modifying the electrical circuit of the hydraulic system and integrating an **Arduino**. ([video](#))
Enabled **wireless communication** for accurate pose commands using a **bluetooth module**.
- **Concordia University** Montréal, Canada
[Research Assistant, AI Robotics](#) Sep 2021 - Dec 2023
Vision-based Object Tracking with two data-driven methods:
 - **Deep Learning:** Introduced an original **real-time** vision-based pose estimation method, demonstrating a **32% reduction in pose error** and a **22% improvement in velocity precision** compared to conventional methods. ([video](#))
 - **Image Feature Optimization:** Optimized a novel image feature function to reduce motion coupling in visual servoing, achieving an **87% improvement in decoupling efficiency**.**Novel Imagery Dataset Generation:** Specific for pose estimation, created a **benchmark dataset** of 600K+ synthetic and real data. Employed DBSCAN for outlier removal. ([video](#))
- **RoboDK** Montréal, Canada
[Intern, Robotic Software Developer](#) May 2022 - Sep 2022
Developed RoboDK software Add-ins:
 - **Real-time IBVS:** Segmenting, and tracking 2D markers using a robot, providing an automation add-in to the software.
 - **3D Reconstruction by Point Cloud Registration**Developed an automatic **bin-filling algorithm** as a base for **object pose estimation** and **bin-picking** algorithms. ([video](#))
- **Chemnitz University of Technology** Chemnitz, Germany
[Intern, Mechanical Design Engineer](#) May 2019 - Jul 2019
Tuned and calibrated the **PID controller** of the tensile fatigue test equipment, significantly improving accuracy in load control and responsiveness. Introduced a novel method for the real-time monitoring of the tensile fatigue test by applying the **Acoustic Emission Sensor** and collecting data. Conducted FEM Analysis of Shaft-Hub Connections, identifying stress distribution and potential failure points. Performed 3D Microscope Imaging and Analysis of crack growth.
- **Tehran Jonoob Technical & Construction Co.** Tehran, Iran
[Mechanical Design Engineer](#) Jun 2017 - Sep 2017
Designed and modelled storage tanks using AutoCAD, ensuring structural integrity and compliance with industry standards.

PROJECTS

- **Concrete Crack Detection via Instance Segmentation**
Developed an instance segmentation model using **YOLOv11-seg** in **PyTorch** to detect cracks in concrete surfaces. Leveraged Roboflow for data preprocessing, augmentation, and training with a public dataset of 5,900+ images. Achieved **mAP@50 of 90.9%**, **precision of 89.2%**, and **recall of 82%**.
- **Simulated Robotic Combat: Path Planning**
Developed a novel **path planning** algorithm in an obstructed environment, demonstrated **faster performance** than established methods. **Implemented an image processing algorithm**, accurately detecting the robots and the obstacles. Specified **dynamic target points** for attacker and defender robots to **optimize combat strategies**.
- **Pick-and-Place to Recycle Objects**
Automating recycling operation using a robotic manipulator in a Gazebo environment. Implementing a **pick-and-place algorithm** by employing **point cloud processing** for object detection and applying **RRT path planning** with real-time ROS integration to achieve collision-free trajectories.

- **Collaborative Pick-and-Pace with Mobile Robot**

Developed a collaborative **pick-and-place** using **ROS** and **Gazebo**:

- First manipulator picks the object from a conveyor belt and places it onto a mobile robot.
- Mobile robot autonomously navigates to deliver objects to a second manipulator.
- Second manipulator picks the object from the mobile robot and places it into a bin.

- **Ball & Beam Controller Design & GUI**

Improving the stability of the ball & beam by implementing linear and **sliding mode** controllers. Designed a **MATLAB GUI** for the ball & beam system, allowing users to set control parameters and initial conditions to animate the experiment, enabling method comparison and easy tuning.

- **3D Design, Printing & FEM Analysis**

CAD design and 3D printing of an industrial equipment replacement part, performing structural analysis using **Abaqus** to validate strength and reliability. **Designed** and **3D printed** a visual marker for pose estimation experiment.

- **Sensor Mount Design & Integration**

Designed an end-effector mount, integrating a camera with a robot arm for vision-based tasks. Developed a custom mount for an acoustic emission sensor for monitoring during part testing.

SKILLS

- **Programming:** Python, C/C++, MATLAB, Git
- **Libraries:** Tensorflow, Keras, Pytorch, Scikit-learn, OpenCV, Open3D, Scikit-image
- **Engineering:** Simulink, PLC, ROS, Arduino, Microcontroller, Kalman Filter
- **CAD:** SOLIDWORKS, AutoCAD, PTC Creo, 3D Printing
- **Robotic Simulation:** RoboDK, Gazebo, CARLA
- **Other:** GNU/Linux, Embedded Systems, Communication Protocol, UDP, UART, SSP, I^2C , PWM

COURSES

- ML Specialization Stanford University
- DL Specialization Deeplearning.ai
- Python for Computer Vision with OpenCV and Deep Learning Udemy
- Self-driving Cars Specialization University of Toronto
- ROS – Robot Operating System TU Delft
- Learn 5 PLCs - Allen Bradley, Siemens, Schneider, Omron & Delta Udemy
- The Arduino Platform and C Programming University of California, Irvine
- Interfacing with the Arduino University of California, Irvine
- Programming for Everybody (Getting Started with Python) University of Michigan
- System Modelling and Simulation with Simulink University of Tehran

REFERENCES

- **Dr. Wen-Fang Xie**, *Professor*

Mechanical Engineering Department, Concordia University, Montréal, Canada

Email: wenfang.xie@concordia.ca

- **Dr. Abolfazl Mohebbi**, *Associate Professor*

Department of Mechanical Engineering, Polytechnique Montréal, Montréal, Canada

Email: abolfazl.mohebbi@polymtl.ca

- **Samuel Bertrand**, *General Manager*

RoboDK, Montréal, Canada

Email: samuel@robodk.com